

BUSINESS ANALYSIS

Reporting & Query Tools and OLAP

10-Mark Answers with Examples & Diagrams

Topics Covered:

1. Reporting and Query Tools and Applications
2. Tool Categories
3. The Need for Applications
4. Cognos Impromptu
5. OLAP – Need
6. Multidimensional Data Model
7. OLAP Guidelines
8. Multidimensional vs Multirelational OLAP
9. Categories of OLAP Tools
10. OLAP Tools and the Internet

1. Reporting and Query Tools and Applications

[10 Marks]

Introduction

Reporting and query tools are front-end software applications that allow business users to access, retrieve, and present data stored in a data warehouse in a meaningful, easy-to-understand format. They form the presentation layer of a data warehousing/business intelligence system, bridging the gap between the technical database structure and the non-technical business user.

Definition

A reporting tool converts raw data into formatted reports (tables, charts, summaries), while a query tool allows users to ask ad-hoc questions of the data using a simplified interface rather than writing SQL directly.

Key Characteristics

- Provide a business-friendly view of underlying data (hide table/column names, joins, keys).
- Allow report scheduling, formatting, and distribution (print, e-mail, web, PDF).
- Support drill-down, sorting, grouping, filtering and simple calculations.
- Enable both pre-built (production) reports and ad-hoc (on-demand) queries.
- Reduce dependency on IT/programmers for routine information needs.

Applications

- Sales performance reports for management review.
- Financial statements and budget-vs-actual reports for finance teams.
- Inventory and stock-level reports for operations.
- Customer segmentation and churn reports for marketing.

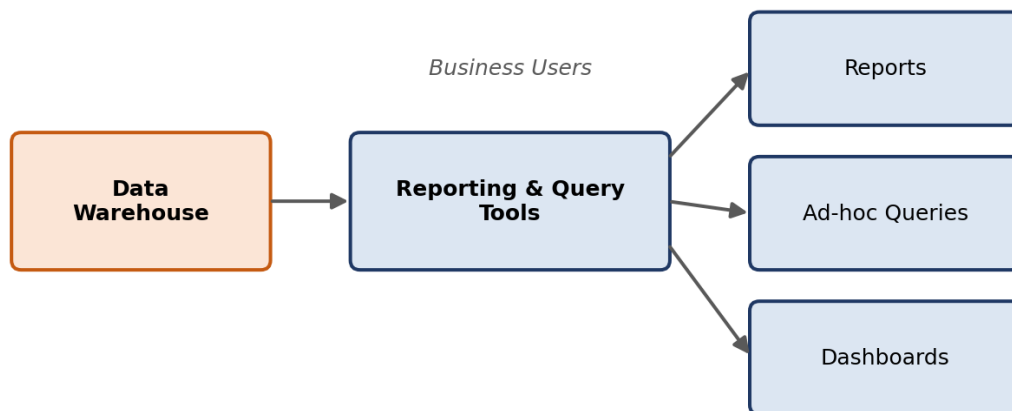


Fig 1.1: Flow from Data Warehouse through Reporting & Query Tools to end-user output

Example:

A retail company's data warehouse stores daily sales transactions. Using a reporting tool, the sales manager generates a weekly 'Sales by Region' report automatically every Monday, while a business analyst uses the same tool's query feature

to ask an ad-hoc question — 'What were total sales for Product X in the North region last quarter?' — without writing any SQL.

Conclusion

Reporting and query tools are essential because they democratize access to warehouse data, allowing decision-makers at all levels to obtain accurate, timely information without technical expertise, thereby improving the speed and quality of business decisions.

2. Tool Categories

[10 Marks]

Introduction

Reporting and query tools used in data warehousing can be classified into distinct categories based on their functionality, target users, and level of sophistication. Understanding these categories helps organizations choose the right tool for the right business need.

Categories of Tools

(a) Reporting Tools

Used to generate formatted, often scheduled reports. Split into: Production reporting tools (used to create regular, high-volume operational reports, e.g., customer invoices, payroll) and Report writers (simple desktop tools like Crystal Reports or MS Access reports used by end users to build their own reports).

(b) Managed Query Tools

Shield end users from the complexity of the database by presenting data through a business-friendly metadata layer (catalog). Users build queries by selecting business terms rather than writing SQL. Example: Cognos Impromptu, Business Objects.

(c) Executive Information System (EIS) Tools

Originally designed to give senior executives a consolidated, high-level view of enterprise performance through dashboards and scorecards. Now used more broadly across all management levels to track KPIs.

(d) OLAP Tools

Provide multidimensional analysis, allowing users to slice, dice, drill down and pivot data across dimensions such as time, product, and region. Example: Cognos PowerPlay, Hyperion Essbase.

(e) Data Mining Tools

Use statistical and machine-learning techniques to discover hidden patterns, trends, and relationships in large datasets that are not obvious through standard querying, e.g., market basket analysis, forecasting.

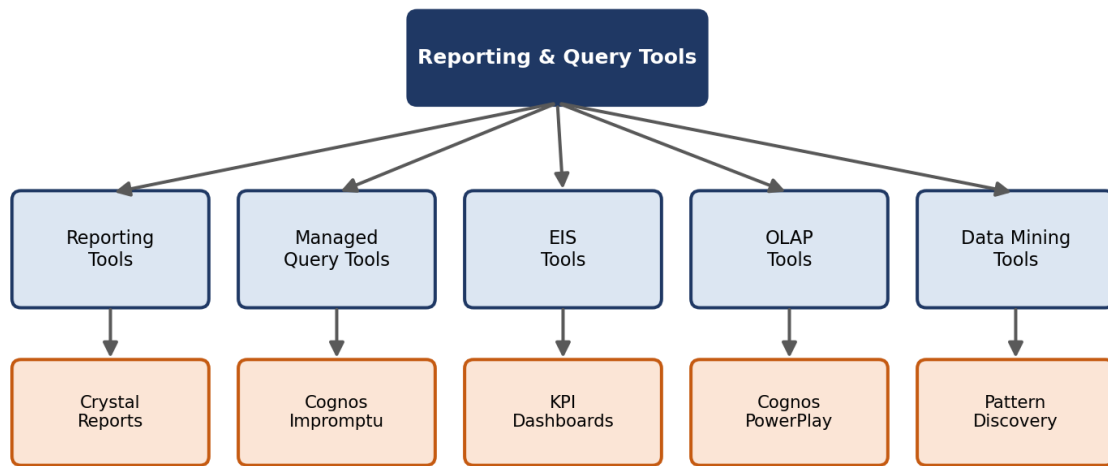


Fig 2.1: Categories of Reporting & Query Tools with representative examples

Example:

A bank may use a report writer for monthly account statements, a managed query tool for branch managers to ask ad-hoc customer queries, an EIS dashboard for the CEO to track overall profitability, an OLAP tool for the finance team to analyze loan performance by branch and quarter, and a data mining tool to detect fraudulent transaction patterns.

Conclusion

Each category serves a different purpose and user skill level; a mature BI environment typically deploys a combination of these categories to serve its diverse user base effectively.

3. The Need for Applications

[10 Marks]

Introduction

While generic reporting and query tools satisfy most information needs, certain business situations demand purpose-built applications rather than open-ended query capability. This need arises because generic tools, though flexible, cannot always deliver analysis in the exact form required for specific, repeated business decisions.

Reasons for the Need

(a) Repetitive and Complex Analysis

Some analyses (e.g., customer lifetime value calculation, risk scoring) are performed frequently and involve complex, multi-step logic. Building this once as a packaged application avoids every user having to reconstruct the logic manually each time.

(b) Presentation in a Usable Form

Standard query tools return raw or tabular results. Certain decisions need results presented in a specific pre-defined, decision-ready format (e.g., a loan-approval scorecard), which a custom application can provide directly.

(c) Consistency Across the Organization

When many users compute the same metric independently using ad-hoc queries, definitions can drift, leading to conflicting numbers. A single, controlled application ensures everyone uses the same calculation logic.

(d) Ease of Use for Non-Technical Users

Applications can be built with guided, wizard-style interfaces so that users with minimal training can perform sophisticated analysis without understanding the underlying data model.

(e) Embedding Business Rules

Applications can enforce organizational business rules, validations, and security automatically, which generic query tools cannot guarantee.

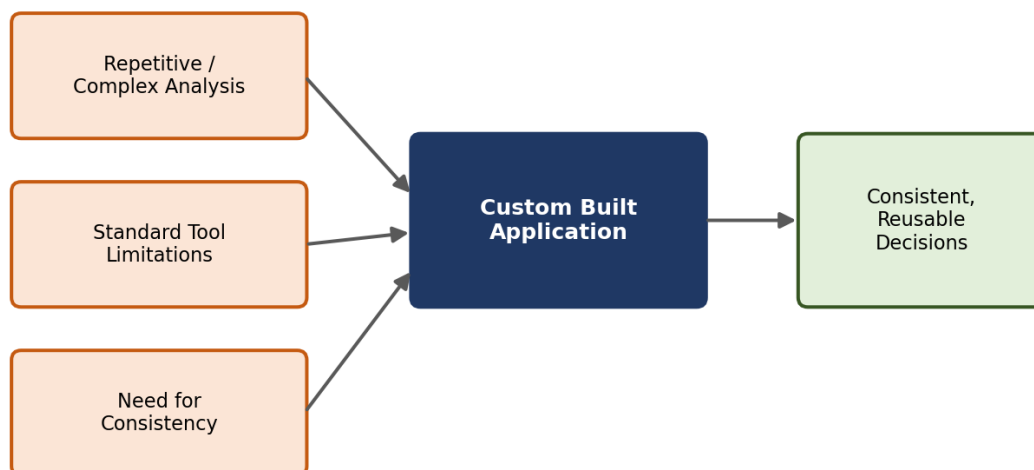


Fig 3.1: Drivers leading to the development of custom analytical applications

Example:

A telecom company needs to calculate 'customer churn risk' every week using a complex formula combining call drop rate, billing delays, and complaint history. Instead of relying on analysts to write this logic repeatedly in a query tool, the company builds a dedicated Churn Prediction Application that automatically applies the same rules and displays a ready-to-use risk score to the retention team.

Conclusion

Custom applications complement generic reporting/query/OLAP tools by delivering consistent, reusable, and decision-ready analytical capability for recurring and complex business problems.

4. Cognos Impromptu

[10 Marks]

Introduction

Cognos Impromptu is a widely used managed query and reporting tool designed to give business users direct, easy access to relational databases without requiring knowledge of SQL or the physical database structure.

Key Features

- Works directly against relational databases; does not require building a separate multidimensional database.
- Uses a Catalog — a metadata layer that maps physical tables and columns into business-friendly folders, headings and terms.
- Allows users to build queries by simply selecting fields from the catalog rather than writing SQL.
- Supports creation of prompts (user-entered filter values), calculated fields, conditional formatting, and grouping.
- Enables report reuse — once a report is built, other users can run it with different filter values.
- Supports drill-down and distribution of reports via print, e-mail, or web publishing.
- Provides governance — administrators control what data and folders each user/group can see through catalog permissions.

Architecture

Impromptu's architecture has three main layers: the underlying relational database, the Catalog that translates technical structures into business terms, and the Impromptu client application through which users build and run reports and queries.

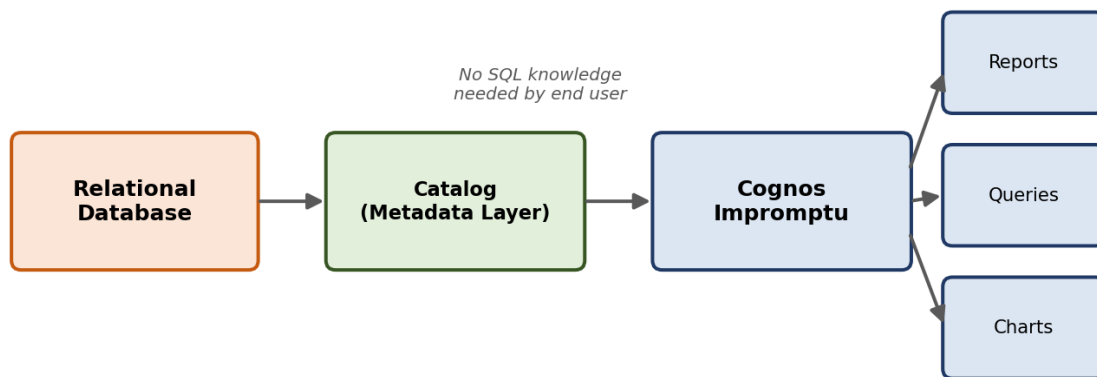


Fig 4.1: Cognos Impromptu architecture — Database, Catalog, and Client output

Example:

In an insurance company, the IT team creates an Impromptu Catalog with folders like 'Policyholders', 'Claims' and 'Premiums', hiding technical table names such as TBL_POL_MSTR. A claims manager, with no SQL knowledge, drags 'Claim Amount' and 'Policy Type' fields into a report to instantly see total claims by policy type for the current year.

Advantages

- Reduces dependency on IT for routine reporting.

- Faster report development compared to hand-written SQL.
- Centralized security and consistent business definitions via the catalog.

Conclusion

Cognos Impromptu exemplifies a managed query tool: by inserting a metadata catalog between the user and the database, it makes enterprise reporting accessible, governed, and consistent for non-technical business users.

5. Online Analytical Processing (OLAP) – Need

[10 Marks]

Introduction

Online Analytical Processing (OLAP) is a technology that enables users to interactively analyze multidimensional data from multiple perspectives. It was developed to overcome the limitations of traditional relational/transactional (OLTP) systems when used for complex business analysis.

Why OLAP is Needed

(a) Limitation of OLTP Systems for Analysis

OLTP systems are optimized for fast, reliable recording of individual transactions (inserts/updates), not for scanning and aggregating millions of rows for analysis. Running complex analytical queries directly on an OLTP database is slow and can disrupt live transaction processing.

(b) Need for Multidimensional Analysis

Business questions are naturally multidimensional — e.g., 'What were sales by product, by region, by quarter?' Relational (2-D) tables require multiple complex joins to answer such questions, whereas OLAP structures data as a cube for direct, fast multidimensional access.

(c) Need for Fast, Interactive Response

Managers need answers in seconds while exploring data interactively (drilling down, slicing). OLAP pre-aggregates data so that responses are near-instant, unlike ad-hoc SQL aggregation on large fact tables.

(d) Need for Trend and Comparative Analysis

OLAP enables comparison across time periods, budget vs actual, and year-over-year growth analysis, which is central to management decision-making but cumbersome in flat relational reporting.

(e) Support for 'What-if' and Ad-hoc Analysis

Business analysts often explore data without a predefined question in mind. OLAP's slice, dice, drill-down and pivot operations support this exploratory style of analysis.

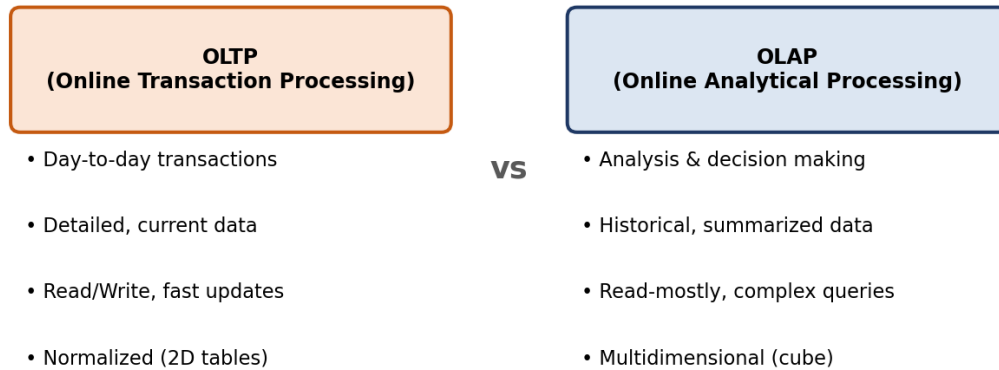


Fig 5.1: OLTP versus OLAP — key differences

Example:

A retail chain's OLTP system records every individual sales transaction across 500 stores. To find out 'which product category grew fastest in the western region over the last 3 years,' the analyst uses an OLAP cube built on top of the warehouse, which returns the answer instantly by aggregating pre-summarized data across the Product, Region and Time dimensions — a query that would take minutes or hours if run directly on the OLTP database.

Conclusion

OLAP is needed because it provides a purpose-built structure and engine for fast, flexible, multidimensional business analysis — a capability that OLTP and simple relational reporting cannot efficiently deliver.

6. Multidimensional Data Model

[10 Marks]

Introduction

The multidimensional data model is the conceptual foundation of OLAP. It organizes data as a 'cube' rather than flat relational tables, allowing data to be viewed and analyzed from multiple perspectives simultaneously.

Core Components

(a) Facts (Measures)

Numeric, quantitative values that are the subject of analysis, such as Sales Revenue, Quantity Sold, or Profit. These sit at the intersection of dimension values inside the cube.

(b) Dimensions

The different perspectives or categories used to analyze facts, such as Time, Product, Region, and Customer. Each dimension provides the axis along which data can be examined.

(c) Hierarchies

Dimensions are organized into hierarchical levels of granularity, e.g., Time: Year → Quarter → Month → Day; Product: Category → Sub-category → Item; Geography: Country → State → City.

(d) The Cube

A cube is formed by the intersection of two or more dimensions with a measure, allowing data to be pre-aggregated and accessed extremely quickly.

Key OLAP Operations on the Model

- Slice – fixing one dimension to a single value to view a 2-D subset of the cube (e.g., sales for Q1 only).
- Dice – selecting a sub-cube by specifying values across two or more dimensions (e.g., Product = 'Electronics' and Region = 'North').
- Drill-down – navigating from a summary level to more detailed data (e.g., Year → Month).
- Roll-up – aggregating detailed data into a higher summary level (e.g., Month → Year).
- Pivot (rotate) – reorienting the cube to view data from a different dimensional perspective.

Multidimensional Data Cube: Sales by Product, Region, Time

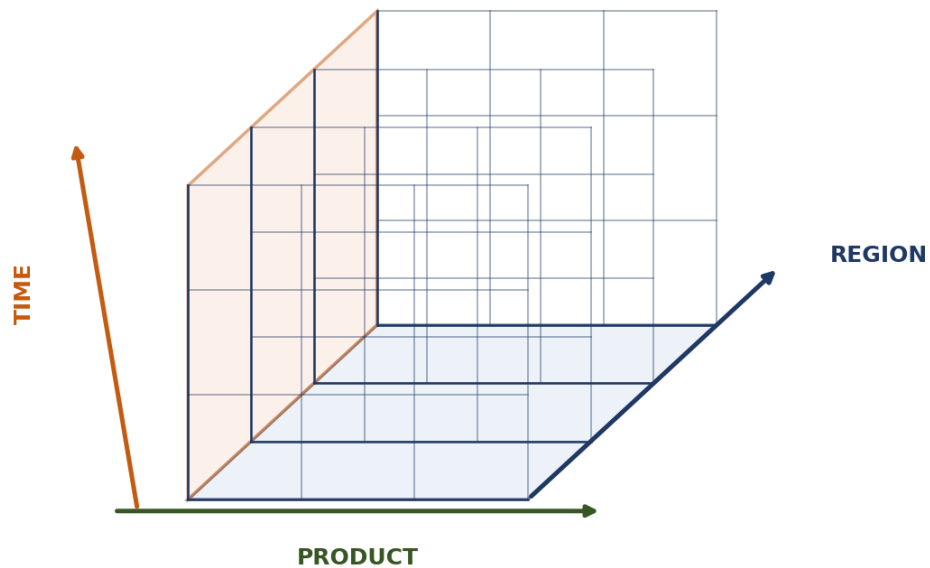


Fig 6.1: A three-dimensional data cube — Product, Region, and Time

Example:

A supermarket chain builds a sales cube with dimensions Product, Store Location and Time, and the measure Sales Amount. A manager can 'slice' the cube to see only December sales, 'dice' it to compare Beverages sales in Delhi and Mumbai, 'drill down' from Year to Month to Day for a suspicious dip, and 'pivot' the view to compare Store Location against Product instead of Time.

Conclusion

The multidimensional data model is what distinguishes OLAP from simple relational reporting — by pre-structuring data as facts and dimensions, it enables fast, intuitive, and flexible analysis of business performance from any angle.

7. OLAP Guidelines

[10 Marks]

Introduction

Dr. E. F. Codd, known as the father of the relational database model, proposed a set of 12 guidelines (rules) in 1993 to define the essential characteristics that a genuine OLAP system should possess. These guidelines are used to evaluate whether a tool can truly be classified as an OLAP product.

Codd's 12 OLAP Guidelines

- 1. Multidimensional Conceptual View – data should be viewed and manipulated as multiple dimensions, matching the way business users naturally think.
- 2. Transparency – the underlying technology, hardware, and data source should be transparent/invisible to the user.
- 3. Accessibility – the OLAP tool should provide a single logical view of data, regardless of where or how it is physically stored (relational, non-relational, etc.).
- 4. Consistent Reporting Performance – performance should not degrade significantly as the number of dimensions or the size of the database increases.
- 5. Client-Server Architecture – the system should operate in a client-server environment, so processing can be distributed for openness and scalability.
- 6. Generic Dimensionality – every dimension should be treated equally in structure and operational capability; no dimension should be favored.
- 7. Dynamic Sparse Matrix Handling – the system should efficiently handle sparse data (many empty cells) without wasting storage or performance.
- 8. Multi-user Support – multiple users should be able to work concurrently on the same or different analytical models with proper security.
- 9. Unrestricted Cross-Dimensional Operations – calculations and operations should be allowed freely across all dimensions.
- 10. Intuitive Data Manipulation – operations like drill-down, slice, dice and pivot should be simple and natural, without needing menus full of complex commands.
- 11. Flexible Reporting – users should be able to arrange rows, columns, and cells in any way needed to analyze data as they wish.
- 12. Unlimited Dimensions and Aggregation Levels – the system should support an unlimited number of dimensions and levels of aggregation (practically limited by hardware, not the software design).

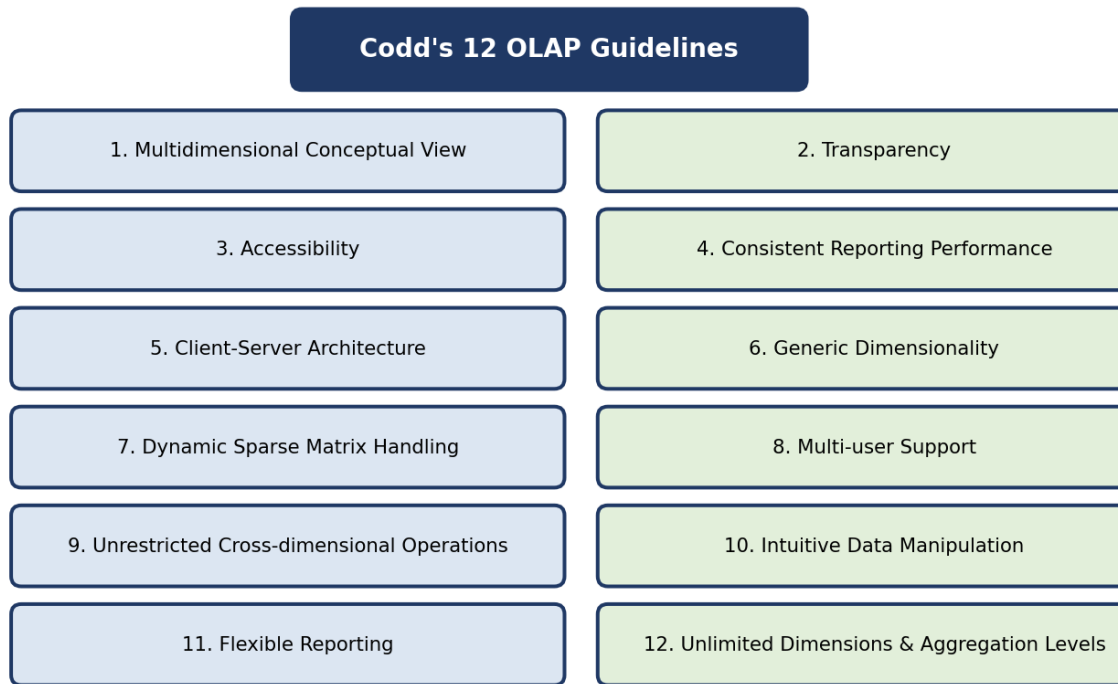


Fig 7.1: Codd's 12 OLAP Guidelines

Example:

When evaluating two OLAP products, an organization checks whether each supports true multidimensional views (Guideline 1), whether adding a new 'Customer Segment' dimension slows down existing reports (Guideline 4), and whether ten analysts can query the same sales cube simultaneously without conflict (Guideline 8) — using Codd's rules as an evaluation checklist.

Conclusion

Codd's guidelines remain the benchmark for assessing OLAP systems; in practice, most commercial tools implement only a subset fully, but the guidelines provide a valuable framework for comparing OLAP products.

8. Multidimensional versus Multirelational OLAP (MOLAP vs ROLAP)

[10 Marks]

Introduction

OLAP systems can be implemented using two fundamentally different storage and processing architectures: Multidimensional OLAP (MOLAP) and Relational/Multirelational OLAP (ROLAP). A third, hybrid approach (HOLAP) combines both.

MOLAP (Multidimensional OLAP)

- Stores data in a proprietary multidimensional cube structure, not standard relational tables.
- Data is pre-calculated and pre-aggregated when the cube is built, giving very fast query response.
- Can suffer from 'data explosion' as the number of dimensions grows, since many dimension combinations must be stored.
- Examples: Hyperion Essbase, Cognos PowerPlay.

ROLAP (Relational/Multirelational OLAP)

- Stores data in standard relational tables using star or snowflake schemas.
- Generates SQL dynamically to compute aggregations at query time rather than pre-storing them.
- Scales better for very large data volumes since it leverages existing RDBMS technology.
- Can be slower than MOLAP for complex queries because aggregation happens on the fly.
- Example: MicroStrategy.

HOLAP (Hybrid OLAP)

Combines both approaches — summary/aggregated data is stored in MOLAP cubes for speed, while detailed data remains in relational tables for scalability, giving a balance of performance and storage efficiency.

Comparison Table

Aspect	MOLAP	ROLAP
Storage	Proprietary cube structure	Relational tables (star/snowflake)
Performance	Very fast (pre-aggregated)	Slower (aggregated at query time)
Scalability	Limited by cube size	Scales well for large volumes
Flexibility	Cube rebuild needed for changes	Flexible, uses existing RDBMS
Examples	Hyperion Essbase, Cognos PowerPlay	MicroStrategy

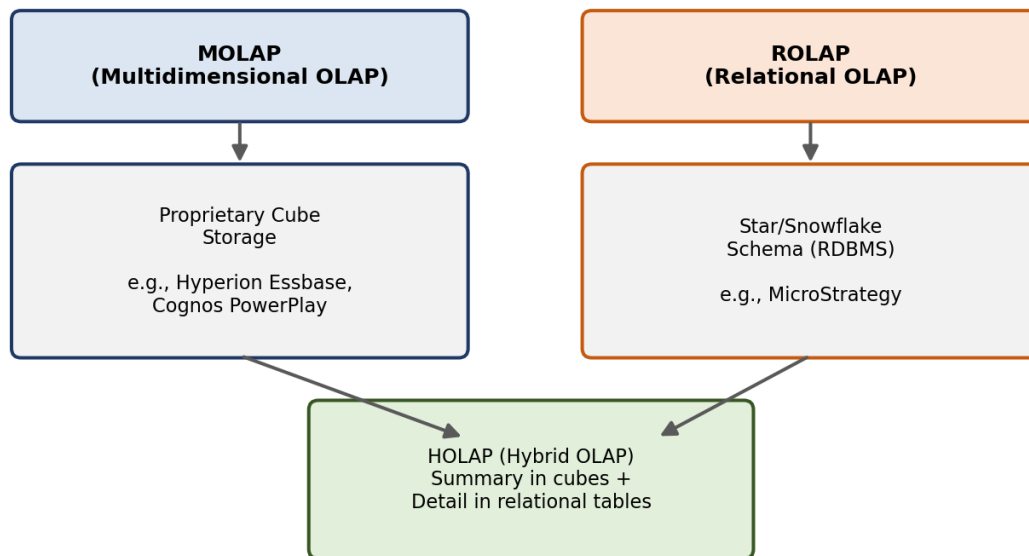


Fig 8.1: MOLAP vs ROLAP architecture, with HOLAP as a hybrid

Example:

A finance department needing sub-second response for a fixed set of monthly budget reports uses a MOLAP cube built in Hyperion Essbase. Meanwhile, the sales department, which has billions of transaction-level records and needs flexibility to query at any granularity, uses a ROLAP tool like MicroStrategy directly on the relational data warehouse.

Conclusion

The choice between MOLAP and ROLAP depends on data volume, query performance needs, and flexibility requirements; many modern OLAP implementations use HOLAP to balance the strengths of both approaches.

9. Categories of (OLAP) Tools

[10 Marks]

Introduction

OLAP tools can be classified into categories based on how they store and process multidimensional data. This classification helps organizations select tools appropriate to their data volume, performance needs, and infrastructure.

Categories

(a) MOLAP Tools

Store data in proprietary multidimensional cubes with pre-calculated aggregations, offering the fastest query performance for moderately sized datasets. Example: Hyperion Essbase, Cognos PowerPlay.

(b) ROLAP Tools

Work directly on relational databases using star/snowflake schemas, generating SQL dynamically. Best suited for very large data volumes. Example: MicroStrategy.

(c) HOLAP Tools

Combine MOLAP and ROLAP — summarized data in cubes, detailed data in relational tables — to balance speed and scalability. Example: Microsoft Analysis Services (in HOLAP mode).

(d) Desktop/Personal OLAP Tools

Lightweight OLAP tools designed for individual analysts to build small, local cubes on their own desktop, often from spreadsheet or extract data, without needing a full enterprise OLAP server. Example: Excel PivotTables with Power Pivot.

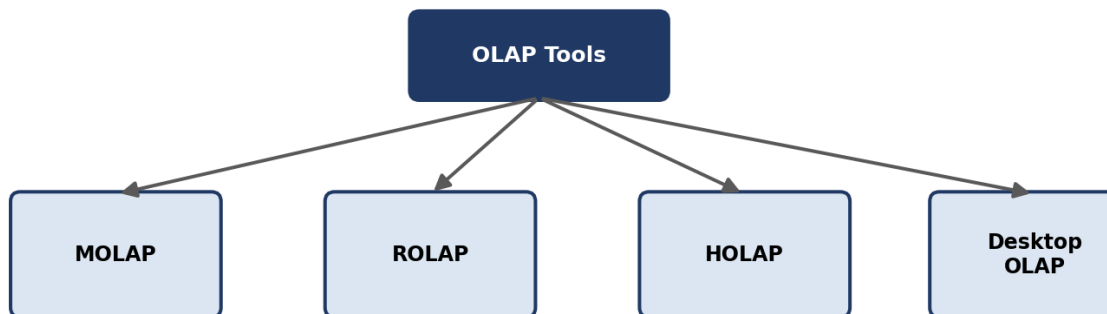


Fig 9.1: Categories of OLAP tools

Example:

A large multinational uses an enterprise MOLAP server for finance reporting, a ROLAP tool for its massive e-commerce clickstream data, HOLAP for its sales data warehouse balancing both needs, and individual regional analysts use Excel Power Pivot as a desktop OLAP tool to build quick ad-hoc cubes from exported data.

Conclusion

The four categories — MOLAP, ROLAP, HOLAP, and Desktop OLAP — together give organizations a spectrum of options ranging from ultra-fast enterprise cubes to lightweight personal analysis tools, allowing the right trade-off between performance, scalability, and cost.

10. OLAP Tools and the Internet

[10 Marks]

Introduction

With the growth of web technologies, OLAP vendors have increasingly moved from heavy desktop ('fat client') applications to web/browser-based front ends, integrating OLAP capability with intranet, extranet, and internet architectures.

Key Developments

(a) Thin-Client / Browser-Based Access

Users can access OLAP cubes and dashboards through a standard web browser instead of installing specialized desktop software, drastically reducing deployment and maintenance costs.

(b) Wider, Distributed Access

Web-enabled OLAP allows access not just within the organization (intranet) but also to external partners, suppliers, and customers (extranet/internet), broadening the reach of analytical insight.

(c) Architecture

A typical web-based OLAP architecture consists of a web browser (client), a web/application server that handles requests and renders results as HTML/JavaScript, an OLAP server (MOLAP/ROLAP engine) that processes multidimensional queries, and the underlying data warehouse.

(d) New Considerations Introduced

- Security – authentication, authorization, and encryption become critical when data is exposed over networks.
- Network performance – large multidimensional result sets must be optimized for transmission over the web.
- Scalability – web architectures must support a much larger and more geographically distributed user base than desktop deployments.
- Standardization – use of open standards (HTML, XML, Java, HTTP/HTTPS) instead of proprietary desktop protocols.

(e) Path to Cloud BI

This shift to browser-based, thin-client OLAP was a key stepping stone toward today's fully cloud-hosted, subscription-based business intelligence and analytics platforms.

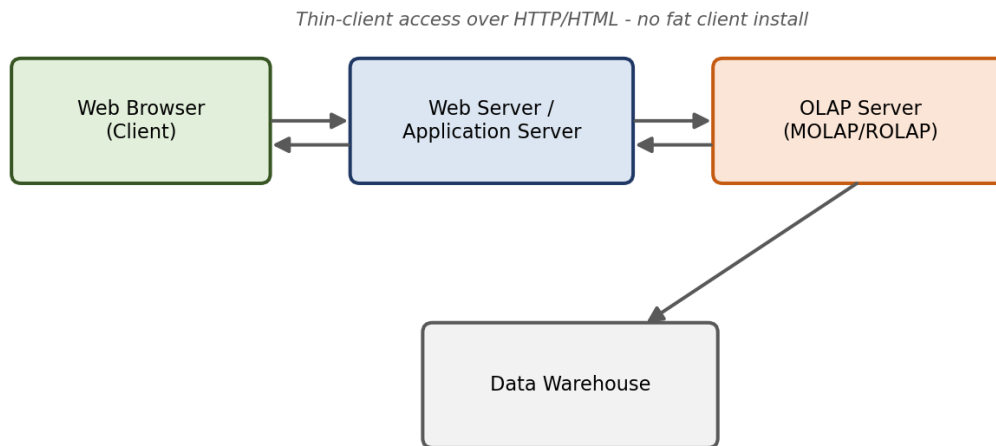


Fig 10.1: Web-based OLAP architecture — browser, web/app server, OLAP server, and data warehouse

Example:

A global logistics company allows its regional partners worldwide to log into a secure web portal from any browser to view shipment performance dashboards built on an OLAP cube, without installing any OLAP client software — replacing an earlier desktop-only OLAP tool that required VPN access and local installation for every user.

Conclusion

Integrating OLAP with internet/web technology transformed analytical tools from isolated desktop applications into accessible, scalable, enterprise-wide (and beyond) decision-support platforms, laying the foundation for modern cloud-based analytics.